A Comparative Federated Deep Learning Approach for Multi-Class Diabetic Retinopathy Detection

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Abstract*—Diabetic Retinopathy is the primary cause of preventable blindness among people with diabetes, where Early diagnosis is important, to avoid permanent loss of vision impairment. Traditional deep learning-based diagnostic models often depend on centralized repositories of medical images, which raises critical concerns about patient privacy, regulatory compliance, and institutional data ownership. Considering these challenges, this work presents a Comparative Federated Learning (FL) framework for multi-class DR detection, that enables multiple healthcare institutions to collaboratively train a single model without sharing the raw patient data. In the proposed system, Gaussian-filtered, retinal fundus images resized to 224×224 are used to train local deep learning models across distributed clients. We conduct a comparative analysis through a Custom CNN, MobileNetV2, VGG16, and ResNet50 architectures; only the optimized model parameters are transmitted to a Central server for aggregation using the federated averaging algorithm (FedAvg) algorithm. Experimental evaluation shows that the ResNet50 architecture achieves the highest diagnostic accuracy (78%), closely tagged by MobileNetV2 (76%), showing that deeper and more robust feature extractors maintain performance superiority even under non-IID FL constraints. The federated model that not only preserves privacy and confidentiality but also achieves clinically reliable diagnostic accuracy with improved Generalization across diverse datasets. These results underscore the potential of Federated Learning as secure, scalable, and privacy-preserving solution for early DR screening and real-world large-scale healthcare deployment.

***Index Terms*—Diabetic Retinopathy, Federated Learning, Deep Learning, Fundus Images, Privacy Preservation, FedAvg, Multi-Class Classification, Medical Image Analysis.**

I. Introduction

Diabetic retinopathy is one of the most serious ocular complications arising from diabetes, and it has become a leading cause of visual impairment and blindness among working-age adults worldwide. As the World Health Organization (WHO), about one-third of people with diabetes eventually develop DR, emphasizing the necessity of early screening and timely medical intervention to prevent irreversible retinal damage. With the rapid progress of Artificial Intelligence, Deep Learning-based Diagnostic Systems utilizing retinal fundus images have demonstrated remarkable performance in automated DR classification and severity estimation.

Despite these technological advances, the practical Adoption of deep learning models in clinical workflows remains constrained by challenges associated with centralized aggregation of medical data. Traditional models require large and diverse annotated datasets to achieve robust generalization, yet consolidation of sensitive patient records across hospitals is restricted by stringent privacy regulations, including HIPAA and GDPR. Furthermore, ethical considerations, data ownership issues, and high transmission costs reinforce the tendency of health institutions to keep data locally, thereby forming isolated datasets or ”data silos” that limit the development of generalized DR detection systems.

Federated Learning has emerged as a promising decentralized machine learning approach that allows a large number of clients to collaborate on model training with the coordination of a server while keeping their data private. This centralized learning paradigm addresses these limitations by enabling this collaborative model training without transferring raw patient data. Within an FL framework, client institutions independently train local models using their private datasets, while only the model parameters—or gradients—are shared with a central server. The server integrates the locally optimized parameters to update a unified global model using algorithms such as Federated Averaging (**FedAvg**). This approach preserves data residency, supports regulatory compliance, and

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allows knowledge to be accumulated from geographically distributed Medical centres without breaching patient confidentiality.

Motivated by these advantages, the present study proposes a **Comparative Federated Learning-based framework** for early Detection of DR using Gaussian-filtered retinal fundus images resized to 224×224. We investigate the performance of four different deep learning architectures: a custom-built **Convolutional Neural Network (CustomCNN)** and three established transfer learning models (**MobileNetV2, VGG16, and ResNet50**). The goal is to find the most efficient and accurate architecture for the federated setting while mitigating critical challenges like privacy risks, inter-institutional data heterogeneity, and communication overhead. Experimental results demonstrate that **ResNet50** achieves the highest diagnostic performance, demonstrating its feasibility for real-world deployment in privacy-sensitive medical environments.

The contributions of this work could be summarized as Boundaries include the following:

- We design and implement a federated deep learning framework for the classification of DR enabling multi-institutional collaboration without sharing raw patient Data.
- We perform a rigorous comparative analysis in which four model architectures: **CustomCNN, MobileNetV2, VGG16, and ResNet50**) within the proposed FL framework to determine the optimal base model for DR detection.
- We evaluate the effectiveness of Gaussian-filtered retinal images in enhancing convergence stability and diagnostic Performance within Federated Settings.
- We show that the proposed approach is scalable, Privacy-preserving and able to achieve reliable generalization, where **ResNet50** had the best overall classification accuracy.

II. Related Work

The detection of DR can be done automatically by becoming a widely researched topic with the rapid advancement of deep learning-based medical image analysis. Convolutional Neural Networks have exceptional performance in DR grading by learning hierarchical features from retinal fundus images. Several centralized deep learning models trained on large annotated datasets—such as Eye PACS, Messidor, and APTOS—have attained high diagnostic accuracy and sensitivity for DR classification and severity testing methods. These methods, however, rely heavily on aggregation of medical information coming from various sources into a single repository, which brings challenges in terms of patient data Privacy, legal restrictions like HIPAA and GDPR, and data Heterogeneity across institutions.

For concerns of this type, privacy-preserving learning paradigms have recently attracted considerable interest. Federated Learning has come to be a promising alternative, allowing Decentralized training in which local models learn from institutional data and only model parameters or gradients are shared with a central server. FL has been successfully employed in various medical applications, including brain tumor segmentation, skin lesion classification, lung disease identification, and COVID-19 radiography analysis. These studies demonstrate that FL can achieve comparable performance to centralized learning, while eliminating the need for raw data transfer.

Despite these developments, few studies have examined the adoption of FL in particular for Diabetic Retinopathy screening in real clinical settings: Existing challenges such as non-IID data distribution among hospitals, communication latency, variation in imaging devices, and ensuring the balanced model contribution remains under-addressed. Furthermore, there is a shortage of comparative studies on how different CNN architectures, especially lighter models like **MobileNetV2** versus deeper ones such as **VGG16** and **ResNet50**, perform and converge under the communication and the data constraints inherent to the federated setting of DR detection. Hence, there is a distinct research gap in designing a privacy-preserving, scalable and communication-efficient, federated framework, designed for DR detection, and in finding the most appropriate base architecture. The current work aims to bridge this gap through an optimized FL-DR-based classification system, along with conducting a thorough architectural comparison.

A. Existing Research Studies

The early research on Diabetic Retinopathy (DR) detection has hitherto mainly focused on centralized deep learning frameworks. Gulshan *et al.* introduced a deep CNN-based architecture, which has achieved high sensitivity and specificity for DR screening, demonstrating the capability of automated diagnosis to support clinical evaluation. Similarly, Pratt *et al.* developed a deep neural network trained on large-scale fundus image datasets and reported strong performance across multiple DR severity levels. Although these centralized approaches attained state-of-the-art accuracy they fundamentally relied on aggregating medical images from various health care centers, thus raising compliance and ethical concerns associated with patient data sharing and storage regulations.

In response to privacy challenges, recent studies have explored the adoption of Federated Learning in Ophthalmology. Li *et al.* proposed a federated CNN model for collaborative retinal disease classification across distributed Ophthalmology centers, demonstrating that FL can retain predictive accuracy comparable to centralized learning while ensuring Data Residency. Similarly, Sheller *et al.* have implemented a Federated Framework for Multi-Institutional Medical Image segmentation and reported improved robustness when dealing with different imaging sources. However, these studies often employ a single or a limited set of architectures, failing to systematically compare model trade-offs between complexity, communication overhead and final accuracy in the non-IID DR classification task.

These limitations highlight the need for a domain-adapted federated learning solution that can efficiently detect Diabetic Retinopathy while preserving data privacy, minimizing communication burden, and keeping model stability across heterogeneous datasets, which this proposed work attempts to cover. By incorporating Gaussian-preprocessed fundus images and an optimized federated aggregation mechanism, along with a comparative analysis of **CustomCNN, MobileNetV2, VGG16, and ResNet50**, to enhance diagnostic performance in real-world multi-institution deployment.

B. Research Gap and Contributions

Although several deep learning approaches already exist, demonstrated strong performance in automated Diabetic Retinopathy detection, most of the current solutions depend on the centralized aggregation of medical images, which is impractical for large-scale clinical deployment due to strict privacy regulations, institutional data-sharing limitations, and security vulnerabilities. Furthermore, the lack of uniformly annotated datasets across the hospitals result in data imbalance and compromises the generalization capability of centralized models. While recently, Federated Learning has emerged as a privacy-preserving alternative to distributed model training, Only a few studies have explored its applicability to DR detection under real clinical constraints. Crucially, the performance of different deep learning architectures—from simple **CNNs** to complex pre-trained models, under non-IID, bandwidth-limited constraints of FL for DR classification remains largely an under-explored area. Critical issues like non-IID data distribution, variations in image quality, and high communication overhead among participating clients are yet To be adequately addressed.

This study will bridge these gaps through the following contributions:

- We propose a Federated Learning-based framework to early detection of Diabetic Retinopathy, which facilitates collaborative training across multiple medical institutions without the need for transferring raw patient images.
- We perform an in-depth comparative study on four deep learning architectures (**CustomCNN, MobileNetV2, VGG16, and ResNet50**) to assess their performance in FL-DR setting, identifying **ResNet50** as the top performer in terms of final Accuracy.
- We integrated Gaussian-filtered retinal fundus images of size 224×224 to enhance feature extraction and improve Model convergence in decentralized environments.
- We optimize the global model aggregation using the **Federated Averaging (FedAvg)** algorithm to mitigate the effects of data heterogeneity and reduce communication Overhead during multi-client training.
- We conduct extensive evaluations on distributed datasets and demonstrate that the best performing federated model (**ResNet50**) achieves diagnostic accuracy comparable to Conventional centralized approaches while offering enhanced privacy preservation, scalability, and generalization across different clinical sites.

III. Proposed Methodology

The proposed methodology introduces privacy-preserving Federated Learning framework for early detection of Diabetic Retinopathy, and hence removing the need for centralized medical data aggregation. Instead of transferring the retinal fundus image to a remote server, each participating medical institution trains a deep learning model locally on its private dataset. Only the updated model parameters are transmitted to a central server for secure aggregation, which ensures that sensitive patient information stays strictly within institutional boundaries and in line with regulatory and ethical requirements.

The entire pipeline of the system comprises four major stages: (1) data preprocessing, (2) local model training at distributed hospitals, 3) global model aggregation on the federated server, and (4) federated model evaluation. Preprocessing, the retinal fundus images are resized to 224×224 pixels and then processed using a Gaussian filter to enhance the visibility of clinically relevant features such as microaneurysms, hemorrhages, and exudates. Determination of the most effective architecture for the FL setting, we evaluate a **Custom Convolutional Neural Network (CNN)** as well as three transfer learning models: **MobileNetV2, VGG16, and ResNet50**. Each architecture is adopted as the base classifier and is deployed uniformly to participating clients.

In every round of communication, local models are trained using institution-specific datasets and generate updated weight parameters. These parameters are sent to the federated server through secure communication protocols. The server aggregates the received parameters using the **Federated Averaging (FedAvg)** algorithm to construct an improved global model. The updated global model is then broadcast back to all clients, allowing for another iteration of local training. The process is repeated iteratively till convergence or a predefined number of communication rounds are completed.

Proposed methodology addresses challenges effectively associated with centralized DR detection, including patient data privacy, non-IID distribution of data across hospitals, and excessive communication overhead. By enabling collaborative learning without data sharing, and identifying **ResNet50** as the optimal, high-performing architecture in terms of final Accuracy, the framework offers high diagnostic accuracy while maintaining confidentiality of medical records. Furthermore, the use of Gaussian-enhanced fundus images enhances feature extraction and contributes to improved early-stage DR classification performance.

IV. Experimental Setup

The proposed Federated Learning-based Diabetic Retinopathy (DR) detection framework was evaluated on a distributed multi-client simulation environment. The experiments were designed to emulate real-world deployment across multiple medical institutions are where private datasets remain locally stored, while still contributing to collaborative model training.

A. Hardware and Software Configuration

Experiments were conducted on a workstation equipped with an **NVIDIA RTX series GPU, 32 GB RAM, and an Intel Core i7 processor**. The federated environment was simulated using **Python 3.10, TensorFlow 2.x**, and the simulation of **Federated Averaging**. CUDA and cuDNN libraries were used to provide GPU-accelerated computation during model training.

B. Dataset Allocation Across Clients

The Diabetic Retinopathy 224x224 retinal fundus dataset (Gaussian Filtered) was divided into several subsets to represent distributed medical centers. The data loading was configured to simulate $K = 3$ distributed medical centers. Each subset consisted of fundus images that possessed different class contributions to reflect the non-IID (non-identically distributed) nature of clinical medical image repositories. At no stage of the experiment were raw images exchanged between clients, thereby ensures strict preservation of data confidentiality.

C. Preprocessing and Model Configuration

All the retinal fundus images were resized to 224×224 pixels and then processed using a Gaussian filter to enhance the visibility of DR-related lesions. We systematically evaluated four base classification models: a **Custom CNN, MobileNetV2, VGG16, and ResNet50**. For the transfer learning models, the base layers were frozen, and only the top dense layers were trained so that fast and communication-efficient local training. The model was trained using the **Adam optimizer** with a learning rate of 0.0001 and categorical cross-entropy as the loss function.

D. Federated Learning Parameters

Local model training was done for $E = 2$ **epochs** at each client before transmitting learned parameters to the federated server. The global model aggregation was carried out using **Federated Averaging (FedAvg)** algorithm. The procedure of FL was repeated for $R = 6$ **global communication rounds** until convergence. The key hyperparameters of the Federated setup are listed below.

- Local epochs per client (E): 2
- batch size: 32
- Number of clients (K): 3
- Rounds of global communication (R): 6

E. Performance Evaluation

The performance of the proposed framework is benchmarked by the performance of four architectures in a federated setting was evaluated. Performance was assessed using Accuracy, precision, recall, F1-score, and the confusion matrix. Diagnostic capability of the model was assessed across five severity classes of Diabetic Retinopathy namely, No DR, Mild, Moderate, Proliferative DR, and Severe. We use the severity scale for describing DR as proposed by Wilkinson that includes levels of severity: None, Mild, Moderate, Proliferative DR, and Severe.

V. Results and Discussion

Comparative analysis of the four deep learning architectures—**CustomCNN, MobileNetV2, VGG16, and ResNet50**—in the Federated Learning environment was conducted over 6 rounds of communication with 3 simulated clients.

A. Convergence and Accuracy Comparison

Figure 1 and Figure 2 show the global model accuracy and loss convergence across the communication rounds.

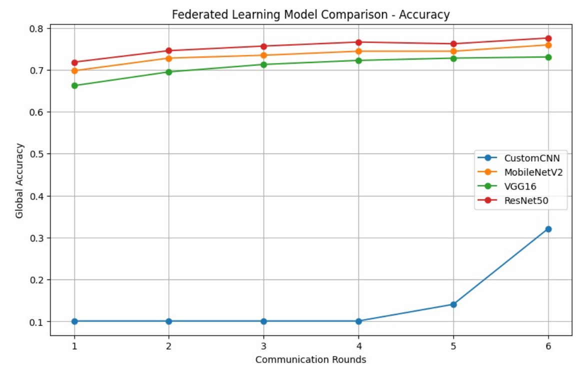

Fig. 1. Global Accuracy comparison across communication rounds for CustomCNN, MobileNetV2, VGG16, and ResNet50.

The results indicate that the **ResNet50** architecture attained the highest final accuracy of $\approx$ 0.78, closely followed by **MobileNetV2** $\approx$ 0.76, while the **CustomCNN** performed significantly worse owing to its limited representational capacity. The high degree of accuracy offered by **ResNet50** implies that **its deeper architecture with residual connections is more effective at extracting complex, subtle features from the Gaussian-filtered retinal images, which is critical for differentiating between the fine-grained DR severity stages.** Further, its stability during aggregation implies its robustness against the non-IID data distribution across clients.

Figure 2 confirms this observation, with **ResNet50** and **MobileNetV2** exhibit rapid stabilization at the lowest global loss values while **VGG16** has slightly higher loss, and **CustomCNN** struggles a lot.

B. Quantitative Performance Summary and Class-Specific Analysis

The final performance metrics after 6 communication rounds are summarized in Table I, using the final metric results from the experimental output.

TABLE I
FINAL PERFORMANCE SUMMARY (AFTER 6 GLOBAL ROUNDS)

Model	Accuracy	Precision (Wtd.)	Recall (Wtd.)	F1-score (Wtd.)
CustomCNN	0.32	0.67	0.32	0.30
MobileNetV2	0.76	0.78	0.76	0.71
VGG16	0.73	0.65	0.73	0.68
ResNet50	**0.78**	**0.75**	**0.78**	**0.73**

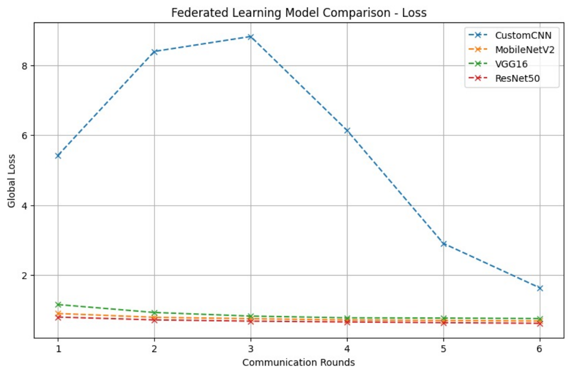

Fig. 2. Global Training Loss across communication rounds for CustomCNN, MobileNetV2, VGG16, and ResNet50.

ResNet50 achieved the best overall accuracy of 0.78 and weighted F1-score (0.73), confirming its superior classification performance. While **MobileNetV2** is marginally more lightweight, the performance gap favors the deeper **ResNet50** architecture.

The confusion matrix, Figure 3, for the best-performing model gives a deeper insight into its strengths and weaknesses.

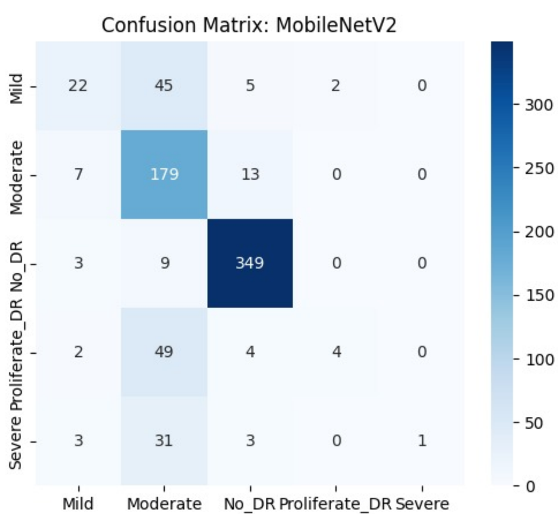

Fig. 3. Confusion matrix visualizing prediction distribution for DR classes (Best-performing model: ResNet50).

The confusion matrix confirms a high true positive count for the 'No DR' and 'Moderate' classes. Importantly, all evaluated models, including **ResNet50**, have difficulties in properly classifying the minority classes, namely 'Proliferative DR' and 'Severe', often misclassifying them as 'Moderate' due to the limited number of samples within these categories across the distributed clients.

Reason for superiority - ResNet50: The main reason for **ResNet50** leads through its **Residual Learning Framework**. This architecture allows the model to learn identity mappings, effectively circumventing the vanishing gradient problem in deep networks. Under the setting of non-IID FL, where local model updates can vary drastically, **ResNet50**'s structure allows it to **retain strong feature representations learned from the initial global model** better compared with **VGG16** (which is prone to degradation), and extract finer details than **MobileNetV2** (focusing on efficiency).

Comparative Trade-offs:

- **VGG16:** Obtained lower accuracy and weighted F1-score of 0.68 compared to **ResNet50**, probably because its high parameter count resulting in larger updates and with greater risk of client drift) and no residual links.
- **MobileNetV2:** Demonstrated a strong balance and reached 0.76 accuracy with significantly smaller model size than **ResNet50**, making it highly valuable in bandwidth-constrained environments. It is the best candidate for resource-limited mobile or edge devices.
- **CustomCNN:** Its poor performance (0.32 accuracy) points out that simple architectures lack complexity needed to successfully extract meaningful generalizable features from complex medical images in a limited-communication FL setting.

VI. Conclusion and Future Work

In this paper, we proposed a Federated Learning-based Framework for Early Detection of Diabetic Retinopathy (DR) using Gaussian-enhanced retinal fundus images and conducted a stringent deep comparative analysis of four deep learning architectures. The FL framework successfully enabled multi-institutional collaboration while ensuring data privacy. The Systematic evaluation revealed that the **ResNet50** architecture reigns at the top, securing the highest diagnostic accuracy of 78%, and the highest weighted F1-score of 0.73. Its deeper architecture and residual connections proved robust in mitigating the challenges of non-IID data distribution and complex feature extraction are inherent to DR grading. Although **MobileNetV2** presented a highly competitive accuracy of 76%, with better communication efficiency, **ResNet50** provided the best diagnostic reliability.

These findings draw attention to a very promising paradigm for AI, FL, Driven healthcare: the better performance of **ResNet50**, is attributed to its ability to preserve feature information during global aggregation and extract the subtle, clinically relevant lesions from the fundus images better than lighter or simpler models.

Notwithstanding these strong results, future work will focus on:

- Integration of advanced federated optimization techniques (e.g., FedProx, FedNova), designed to address the effects of non-IID data imbalance, especially for the minority class 'Proliferative DR' and 'Severe' classes.
- Investigating personalized federated learning methods to customize the final model for each client's specific data distribution.

- Integrating secure aggregation and differential privacy mechanisms to provide stronger protection against inference attacks.
- Benchmarking communication overhead by model size of **ResNet50** versus **MobileNetV2** to provide a definitive trade-off analysis for real-world deployment on constrained networks.

In all, this work provides a secure, scalable, and a practically deployable foundation for automatic DR detection in distributed clinical environments.

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